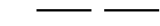
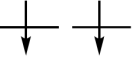
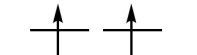
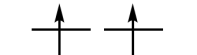


Atomic Structure/Periodicity

- Which species has the largest ionic radius?
(A) F^- (B) Na^+ (C) Mg^{2+} (D) Al^{3+}
- Which set of quantum numbers is not possible for an electron?
(A) $n = 1, l = 1, m_l = 0, m_s = +1/2$
(B) $n = 2, l = 1, m_l = 1, m_s = -1/2$
(C) $n = 3, l = 2, m_l = -2, m_s = +1/2$
(D) $n = 4, l = 0, m_l = 0, m_s = +1/2$
- For which gas-phase atoms are the 4s orbitals lower in energy than the 3d orbitals?
I. H II. K
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II
- Which element has the largest first ionization energy?
(A) F (B) Ne (C) Cl (D) Ar
- Which best describes the distribution of electrons among the 3d orbitals in $Fe(H_2O)_6^{2+}$ in its most stable state?
(A)  (B) 
(C)  (D) 
(A)  (D) 
(A)  (D) 
- An isotope of phosphorus undergoes β^- decay to give which daughter nucleus?
(A) ^{27}Al (B) ^{29}Si (C) ^{31}P (D) ^{33}S
- Which atom has the smallest first ionization energy?
(A) Mg (B) Al (C) Si (D) P
- How many electrons in a ground-state atom of uranium (U) have the principal quantum number $n = 4$?
(A) 4 (B) 16 (C) 18 (D) 32
- Which property of a photon is directly proportional to its frequency?
(A) Energy (B) Momentum
(C) Velocity (D) Wavelength
- Photoelectron spectroscopy of Ar provides information about which properties of Ar?
I. Ionization energy II. Electronic subshell structure
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II
- Which statements about fluorine are correct?
I. It has the highest electronegativity of any element.
II. Its electron affinity is the largest in magnitude of any element.
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II
- Thorium-232 decays by emission first of an alpha particle, then of a beta particle, and then of a second alpha particle. What is the daughter nucleus after the fourth disintegration?
(A) ^{228}Th (B) ^{228}Ac
(C) ^{224}Ra (D) ^{224}Rn
- Which element's electronegativity is most similar to that of sulfur (S)?
(A) O (B) F (C) As (D) Se
- Two atoms of the same element, **A** and **B**, are in identical excited states. Atom **A** emits a photon of red light, while atom **B** emits a photon of blue light. Which statement best describes the energies of the two atoms after the emission of light?
(A) Atom **A** is in a lower-energy state than atom **B**.
(B) Atom **A** is in a higher-energy state than atom **B**.
(C) Atom **A** and atom **B** have the same energy.
(D) The relative energies of **A** and **B** cannot be determined from the information given.
- Two orbitals in an atom are both spherically symmetric but differ in size. Which statements about the orbitals' quantum numbers must be correct?
I. The two orbitals have different values of n .
II. The two orbitals have the same value of m_l .
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II
- Which is a pair of gas-phase species with the same ground state electron configuration?
(A) V and Cr^+ (B) Mn^+ and Fe^-
(C) Ni and Pd (D) Zn and Ga^+

Atomic Structure/Periodicity

17. Which list arranges from highest to lowest the standard reduction potentials E° for reduction of the divalent group 10 metal ions $M^{2+}(aq)$ to $M(s)$?
- (A) $Ni > Pd > Pt$ (B) $Pt > Pd > Ni$
(C) $Pd > Ni > Pt$ (D) $Pd > Pt > Ni$
18. A nucleus absorbs a neutron and undergoes fission to produce $^{103}_{40}Zr$, $^{134}_{54}Xe$, and three additional neutrons. What is the fissile nucleus?
- (A) ^{235}U (B) ^{237}U (C) ^{237}Pu (D) ^{239}Pu
19. Which species has the largest ionic radius?
- (A) S^{2-} (B) Cl^- (C) K^+ (D) Ca^{2+}
20. Which of the following statements regarding ionization energy of gas-phase atoms is correct?
- (A) The first ionization energy of a group 1 element is always greater than the first ionization energy of the group 2 element in the same row of the periodic table.
(B) In the second row of the periodic table, the first ionization energy is directly proportional to the atomic radius.
(C) Among the elements from Al ($Z = 13$) to Ar ($Z = 18$), the first ionization energy increases monotonically with atomic number.
(D) Among the elements of group 16, the first ionization energy decreases monotonically with atomic number.
21. How many electrons in a ground state gas-phase Al atom have the quantum number $l = 1$?
- (A) 1 (B) 3 (C) 5 (D) 7
22. What is the ground-state electron configuration of a gas-phase Mn^{2+} ion?
- (A) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$ (B) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$
(C) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7$ (D) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^2$
23. What wavelengths of light are capable of breaking a bond with a bond strength of 425 kJ mol^{-1} ?
- (A) Only $\lambda < 281 \text{ nm}$ (B) Only $\lambda > 281 \text{ nm}$
(C) Only $\lambda < 355 \text{ nm}$ (D) Only $\lambda > 355 \text{ nm}$
24. When ^{105}Te decays by alpha emission, how many protons are in the daughter nuclide that is produced?
- (A) 48 (B) 50 (C) 52 (D) 53
25. Which statement best describes the amount of energy released by gas-phase oxygen and sulfur atoms upon accepting an electron (electron affinities)?
- (A) O releases less energy than S because the smaller size of O causes greater electron-electron repulsion.
(B) O releases less energy than S because O has a lower mass than S.
(C) O releases more energy than S because the electron enters a $2p$ orbital in O but a $3p$ orbital in S.
(D) O releases more energy than S because O has a larger effective nuclear charge Z_{eff} than S.
26. Which element has chemical properties most similar to those of S?
- (A) O (B) P (C) Cl (D) Se
27. What is the largest number of electrons that can have both a principal quantum number $n = 4$ and an orbital angular momentum quantum number $l = 2$ in a given atom?
- (A) 1 (B) 3 (C) 5 (D) 10
28. Which best represents the arrangement of electrons in a ground-state gas-phase Ni^{2+} ion?
- $4s$ $3d$
- (A) $[Ar] (\uparrow\downarrow) (\uparrow\downarrow)(\uparrow\downarrow)(\uparrow\downarrow) () ()$
(B) $[Ar] (\uparrow\downarrow) (\uparrow\downarrow)(\uparrow) (\uparrow) (\uparrow) (\uparrow) (\uparrow)$
(C) $[Ar] () (\uparrow\downarrow)(\uparrow\downarrow)(\uparrow\downarrow)(\uparrow\downarrow) () ()$
(D) $[Ar] () (\uparrow\downarrow)(\uparrow\downarrow)(\uparrow\downarrow)(\uparrow) (\uparrow)$
29. What is the wavelength of a photon whose energy is 334 kJ mol^{-1} ?
- (A) 222 nm (B) 358 nm
(C) 590 nm (D) 2994 nm
30. What nuclide decays by alpha emission to form ^{219}Rn ?
- (A) ^{215}Po (B) ^{219}At (C) ^{219}Fr (D) ^{223}Ra
31. Photons of what wavelength of light have an energy of $1.00 \times 10^3 \text{ kJ mol}^{-1}$?
- (A) $2.00 \times 10^{31} \text{ m}$ (B) $1.20 \times 10^{-7} \text{ m}$
(C) $1.20 \times 10^{-4} \text{ m}$ (D) $8.36 \times 10^6 \text{ m}$
32. In which set are all three species isoelectronic?
- (A) H^+ , Li^+ , Na^+ (B) C , N^- , O^{2-}
(C) Se^{2-} , Br^- , Rb^+ (D) I , Xe , Cs

Atomic Structure/Periodicity

33. Which change in principal quantum number n involves the greatest change in energy?
 (A) $n = 1 \rightarrow n = 2$ in H (B) $n = 2 \rightarrow n = 3$ in He^+
 (C) $n = 3 \rightarrow n = 4$ in Li^{2+} (D) $n = 4 \rightarrow n = 5$ in Be^{3+}
34. Which gas-phase ion has the greatest number of unpaired electrons in its ground state?
 (A) Fe^{3+} (B) Co^{3+} (C) Ni^{3+} (D) Cu^{3+}
35. Which principle restricts the occupancy of an atomic orbital to no more than two electrons?
 (A) The principle of conservation of energy
 (B) The principle of conservation of angular momentum
 (C) The Pauli exclusion principle
 (D) The Heisenberg uncertainty principle
36. When copper-64 decays by positron emission, what daughter nuclide is formed?
 (A) ^{64}Ni (B) ^{65}Ni (C) ^{64}Zn (D) ^{65}Zn
37. Which is a possible set of quantum numbers n, l, m_l, m_s for a valence electron of sulfur (S)?
 (A) 3, 2, 2, $1/2$ (B) 3, 1, -1, $1/2$
 (C) 2, 1, 0, $1/2$ (D) 3, 2, 0, $-1/2$
38. Which period 3 element has these successive ionization energies (in kJ mol^{-1})?

IE_1	IE_2	IE_3	IE_4
786.5	1577.1	3231.6	4355.5
IE_5	IE_6	IE_7	
16091	19805	23780	

- (A) Al (B) Si (C) P (D) S
39. Which is the ground-state electron configuration of gas-phase Co^{2+} ?
 (A) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$
 (B) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^5$
 (C) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 4d^5$
 (D) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7$
40. A fluorescent dye absorbs a photon of light of 485 nm and emits a photon of light at 540 nm. How much energy is lost as heat?
 (A) $4.17 \times 10^{-20} \text{ J}$ (B) $3.68 \times 10^{-19} \text{ J}$
 (C) $4.10 \times 10^{-19} \text{ J}$ (D) $3.06 \times 10^{-18} \text{ J}$

41. Which gas-phase atom releases the most energy when an electron is added to it?
 (A) Na (B) Cl (C) K (D) Br
42. The isotope ^{44}Ti undergoes electron capture. Which daughter isotope is produced?
 (A) ^{40}K (B) ^{40}Ca (C) ^{44}Sc (D) ^{44}V
43. For which element is the +2 oxidation state LEAST common?
 (A) Al (B) V (C) Fe (D) Sm
44. How many completely filled p orbitals are present in a ground state Cl atom?
 (A) 1 (B) 2 (C) 4 (D) 5
45. Which atom has the highest first ionization energy?
 (A) N (B) O (C) P (D) S
46. The electronegativity of gallium (1.8) is greater than that of aluminum (1.6). Which is the best explanation for this difference?
 (A) Ga is larger than Al, so its valence electrons experience less electron-electron repulsion.
 (B) Ga has more protons than Al, so its valence electrons are more attracted to the nucleus.
 (C) Ga has a filled $3d$ subshell, whose electrons incompletely screen the nucleus.
 (D) Ga experiences significant relativistic effects, which contract the valence orbitals.
47. Which set of quantum numbers could NOT correspond to an electron in a ground-state gas phase Pd atom?
 (A) $n = 2, l = 1, m_l = -1, m_s = 1/2$
 (B) $n = 3, l = 3, m_l = -1, m_s = -1/2$
 (C) $n = 4, l = 0, m_l = 0, m_s = -1/2$
 (D) $n = 4, l = 2, m_l = 2, m_s = 1/2$
48. A certain nuclide undergoes beta decay to form ^{70}Ge . If the same nuclide undergoes electron capture instead, what daughter nuclide forms?
 (A) ^{70}Zn (B) ^{70}Ga (C) ^{70}Ge (D) ^{70}As

Atomic Structure/Periodicity

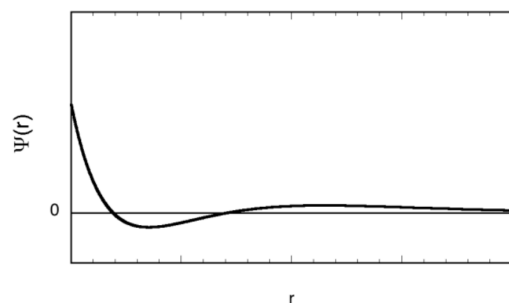
49. Which statement regarding the relative energies of monochromatic light with $\lambda = 800$ nm and monochromatic light with $\lambda = 400$ nm is correct?
- (A) 800 nm light has half as much energy per mole of photons as 400 nm light.
(B) 800 nm light has the same energy per mole of photons as 400 nm light.
(C) 800 nm light has twice as much energy per mole of photons as 400 nm light.
(D) No conclusion may be drawn regarding the relative energy per mole of photons without knowing the intensity of the light.
50. How many electrons in a ground-state As atom in the gas phase have quantum numbers $n = 3$ and $l = 1$?
- (A) 1 (B) 2 (C) 3 (D) 6
51. In which list are the elements in order of increasing first ionization energy?
- (A) Ca < Si < P < N (B) N < P < Si < Ca
(C) Ca < N < P < Si (D) N < Si < P < Ca
52. What mode of radioactive decay does ^{66}Cu undergo?
- (A) Alpha decay (B) Beta decay
(C) Electron capture (D) Positron emission
53. Which gas-phase atoms in their ground states are diamagnetic?
- I. Fe
II. Zn
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
54. The boiling points of the alkali metals decrease from lithium to cesium. Which trend is most closely related to the decreasing boiling points?
- (A) Increasing atomic radius down the group
(B) Increasing atomic mass down the group
(C) Decreasing electronegativity down the group
(D) Decreasing electron affinity down the group
55. What is the electron configuration of the Al^{3+} ion?
- (A) $1s^2 2s^2 2p^6$ (B) $1s^2 2s^2 2p^6 3s^2 3p^1$
(C) $1s^2 2s^2 2p^6 3s^2 3p^4$ (D) $1s^2 2s^2 2p^6 3s^2 3p^6$
56. Which metal has the highest melting point?
- (A) K (B) Ca (C) Fe (D) Zn
57. The wavelength of one line in the emission spectrum of C is 538 nm. What is the energy of one photon with this wavelength?
- (A) 3.69×10^{-19} J (B) 3.69×10^{-26} J
(C) 3.56×10^{-40} J (D) 1.19×10^{-48} J
58. An orbital in a ground-state gas-phase As atom has $n = 3$, $l = 1$. How many electrons are in this orbital?
- (A) 0 (B) 1 (C) 2 (D) 6
59. Atoms of which element have the smallest ionization energy?
- (A) Na (B) F (C) K (D) Cl
60. The acidity of the group 16 hydrides increase going down the group ($\text{H}_2\text{O} \ll \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$). Which is the best explanation for this trend?
- (A) The electronegativity of the group 16 elements increases going down the group.
(B) The polarizability of the group 16 elements increases going down the group.
(C) The polarity of the X–H bond increases going down the group.
(D) The H–X–H bond angle increases going down the group.
61. How many total orbitals are there with principal quantum number $n = 4$?
- (A) 1 (B) 4 (C) 9 (D) 16
62. Addition of an electron to a gas-phase Si atom results in the release of more energy than addition of an electron to a gas-phase P atom. What is the best explanation for their relative electron affinities?
- (A) The electron added to Si experiences less electron-electron repulsion than the electron added to P.
(B) The electron added to Si enters a lower-energy subshell than the electron added to P.
(C) Si is more electronegative than P.
(D) Si is smaller than P.
63. The wavelength of one of the spectral lines of helium is 492 nm. What is the energy of a photon with this wavelength?
- (A) 3.26×10^{-40} J (B) 3.26×10^{-31} J
(C) 4.04×10^{-28} J (D) 4.04×10^{-19} J
64. Which of the following gas-phase ions has the largest number of unpaired electrons in its ground state?
- (A) Cr^{3+} (B) Co^{3+} (C) Ni^{2+} (D) Cu^{2+}

Atomic Structure/Periodicity

⁵²Cr undergoes radioactive decay to give ⁵²Cr by what decay mode?

- (A) Alpha emission (B) Beta emission
(C) Positron emission (D) Gamma emission

66. An orbital has the radial wavefunction shown below. What orbital is it?



- (A) 1s (B) 2s (C) 3s (D) 4s

67. Which of the following statements is best supported by the data from Rutherford's experiment of scattering alpha particles with a thin metal foil?

- (A) The mass and positive charge of an atom are concentrated in its center.
(B) Electrons in atoms occupy only certain specific energy levels.
(C) Moving particles can also be described as waves.
(D) Atoms of a given element do not all have identical masses.

68. What is the formula of the most stable oxide of francium?

- (A) Fr₂O (B) FrO (C) Fr₂O₃ (D) FrO₂

69. As atomic number increases from 11 to 15, the atomic radii of the elements

- (A) increase
(B) decrease
(C) increase, then decrease
(D) decrease, then increase

70. Nitrogen has a greater first ionization energy than oxygen. What is the best explanation for this observation?

- (A) Nitrogen is more electronegative than oxygen.
(B) A nitrogen atom is smaller than an oxygen atom.
(C) The electron ionized from nitrogen experiences less electron-electron repulsion than the electron ionized from oxygen.
(D) The electron ionized from nitrogen is a 2s electron, while the electron ionized from oxygen is a 2p electron.

71. Which set of quantum numbers n, l, m_l, m_s is invalid?

- (A) 1, 1, 0, $-\frac{1}{2}$ (B) 2, 0, 0, $+\frac{1}{2}$
(C) 3, 1, 0, $+\frac{1}{2}$ (D) 4, 3, 2, $-\frac{1}{2}$

72. What is the product of alpha emission from the isotope uranium-238?

- (A) ²³²Th (B) ²³⁴Th (C) ²³⁷Np (D) ²³¹Pa

73. What is the value of the quantum number l for a 5p orbital?

- (A) 1 (B) 2 (C) 3 (D) 4

74. Which element has chemical properties most similar to those of P?

- (A) N (B) Al (C) S (D) As

75. Which metal has the lowest melting point?

- (A) Li (B) Na (C) K (D) Rb

76. Which gas-phase atom has no unpaired electrons in its ground state?

- (A) Li (B) Be (C) B (D) C

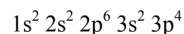
77. Which halogen atom has the greatest electron affinity?

- (A) F (B) Cl (C) Br (D) I

78. Which electronic transition in atomic hydrogen corresponds to the emission of visible light?

- (A) $n = 5 \rightarrow n = 2$ (B) $n = 1 \rightarrow n = 2$
(C) $n = 3 \rightarrow n = 4$ (D) $n = 3 \rightarrow n = 1$

79. A sulfur atom in its ground state has the electron configuration



How many orbitals are occupied by at least one electron?

- (A) 3 (B) 5 (C) 8 (D) 9

Atomic Structure/Periodicity

80. For which of the following transitions would a hydrogen atom absorb a photon with longest wavelength?

(A) $n = 1$ to $n = 2$ (B) $n = 3$ to $n = 2$
(C) $n = 5$ to $n = 6$ (D) $n = 7$ to $n = 6$

81. Which set of quantum numbers is *not* possible?

(A) $n = 2, l = 1, m_l = +1, m_s = -1/2$
(B) $n = 3, l = 2, m_l = +1, m_s = +1/2$
(C) $n = 4, l = 4, m_l = -1, m_s = +1/2$
(D) $n = 5, l = 2, m_l = 2, m_s = -1/2$

82. What property of the oxygen atom is represented by the equation $O(g) + e^- \rightarrow O^-(g)$?

(A) electronegativity (B) first electron affinity
(C) first ionization energy (D) lattice energy

83. Which of the following isoelectronic species has the largest radius?

(A) K^+ (B) Ca^{2+} (C) P^{3-} (D) S^{2-}

84. Which group best illustrates the transition from non-metallic to metallic behavior with increasing atomic number?

(A) Be, Mg, Ca, Sr (B) N, P, As, Sb
(C) F, Cl, Br, I (D) Fe, Ru, Os, Hs

85. To whom is the discovery of the nuclear atom attributed?

(A) Neils Bohr (B) Louis deBroglie
(C) Robert Millikan (D) Ernest Rutherford

86. Each of the following atomic orbitals is possible except

(A) $1s$. (B) $2p$. (C) $3f$. (D) $4d$.

87. The ion $^{55}\text{Mn}^{2+}$ contains which combination of protons, neutrons and electrons?

	protons	neutrons	electrons
A	25	30	23
B	25	55	23
C	27	30	25
D	30	25	28

(A) A (B) B (C) C (D) D

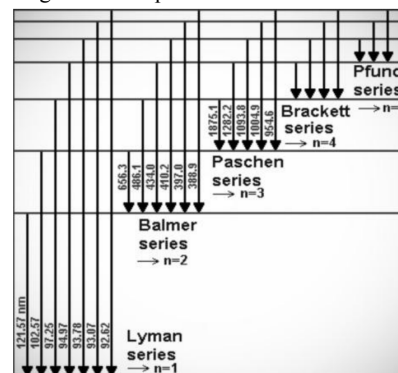
88. Which atom has the highest electronegativity?

(A) Na (B) P (C) Cl (D) Br

89. In which set are both elements metalloids?

(A) Cr and Mo (B) Ge and As
(C) Sn and Pb (D) Se and Br

90. Which spectral series for hydrogen contains lines in the visible region of the spectrum?



(A) Lyman series (B) Balmer series
(C) Paschen series (D) Brackett series

91. The isotope ^{14}C undergoes radioactive decay slowly. Which mode of decay is most likely?

(A) alpha emission (B) beta emission
(C) positron emission (D) electron capture

92. The quantum numbers, $n = 4, l = 1, m_l = 0$ could represent a valence electron in which atom in its ground state?

(A) Fe (B) In (C) Pd (D) Se

93. A gas phase atom with the electron configuration

$$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$$

loses three electrons. What is the electron configuration of the resulting gas phase ion?

(A) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$
(B) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^4$
(C) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$
(D) $1s^2 2s^2 2p^6 3s^2 3p^5 4s^1 3d^5$

94. Which existing element would the chemistry of element 119 most resemble?

(A) Rn ($Z = 86$) (B) Fr ($Z = 87$)
(C) Ra ($Z = 88$) (D) Ac ($Z = 89$)

95. Which atom has the largest atomic radius?

(A) S (B) Cl (C) Se (D) Br

Atomic Structure/Periodicity

96. Of the elements listed, which has the highest first ionization energy?
(A) Li (B) Be (C) Na (D) Mg
97. Which element has atoms with exactly four valence electrons in its ground state?
(A) Ca (B) Cr (C) Si (D) S
98. Gas-phase atoms of which element have an occupied 5d orbital in their ground state?
(A) Ag ($Z = 47$) (B) Ba ($Z = 56$)
(C) Eu ($Z = 63$) (D) Ir ($Z = 77$)
99. Which gas-phase atom in its ground state could have an electron with the quantum numbers: $n = 3$, $l = 2$, $m_l = 0$, $m_s = -1/2$?
(A) Na (B) Mg (C) P (D) Ti
100. Properties of the alkaline earth metals that increase from Be to Ba include which of the following?
I Atomic radius II Ionization energy III Nuclear charge
(A) I and II only (B) I and III only
(C) II and III only (D) I, II, and III
101. In which list are the ions arranged in order of decreasing size?
(A) S^{2-} , Br^- , K^+ , Ca^{2+} (B) Br^- , S^{2-} , K^+ , Ca^{2+}
(C) K^+ , Ca^{2+} , S^{2-} , Br^- (D) Ca^{2+} , K^+ , S^{2-} , Br^-
102. The element Lawrencium was first synthesized by the reaction: ${}^{252}_{98}\text{Cf} + {}^{11}_5\text{B} \rightarrow {}^{258}_{103}\text{Lr} + \text{---}$
What products are needed to balance this equation?
(A) $4\alpha + {}^1_0\text{n}$ (B) $5\text{ }^0_{-1}\text{e}$
(C) $5\text{ }^0_1\text{e}$ (D) $5\text{ }^1_0\text{n}$
103. Which statement concerning visible light is correct?
(A) The product of wavelength and frequency is a constant for visible light in a vacuum.
(B) As the wavelength of light increases the energy of a photon increases.
(C) As the wavelength of light increases its amplitude also increases.
(D) Green light has a higher frequency than blue light.
104. Ernest Rutherford's scattering experiment demonstrated the existence of the
(A) alpha particle. (B) electron.
(C) neutron. (D) nucleus.
105. The removal of an electron from which gaseous atom requires the greatest amount of energy?
(A) Na (B) Cl (C) K (D) Br
106. For which pair of species is the difference in radii the greatest?
(A) Li and F (B) Li^+ and F^-
(C) Li^+ and O^{2-} (D) O^{2-} and F^-
107. The electron configuration of cobalt ($Z = 27$) is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^2$. How many unpaired electrons are in a gaseous Co^{3+} ion in its ground state?
(A) 0 (B) 2 (C) 4 (D) 6
108. Which element has the greatest electrical conductivity?
(A) As (B) Ge (C) P (D) Sn
109. Which properties of electromagnetic radiation are inversely related?
(A) amplitude and frequency
(B) energy and wavelength
(C) energy and frequency
(D) wavelength and amplitude
110. Which electronic transition in a hydrogen atom releases the greatest amount of energy?
(A) $n = 3 \rightarrow n = 2$ (B) $n = 5 \rightarrow n = 3$
(C) $n = 6 \rightarrow n = 5$ (D) $n = 3 \rightarrow n = 6$
111. Which must represent an atom in an excited state?
(A) $1s^2 2s^2 2p^1$ (B) $1s^2 2s^2 2p^2$
(C) $1s^2 2s^2 2p^2 3s^1$ (D) $1s^2 2s^2 2p^5$
112. Which quantum numbers represent the orbitals being filled in the ground state for the elements Sc (21) to Zn (30)?
(A) $n = 3$, $l = 1$ (B) $n = 3$, $l = 2$
(C) $n = 4$, $l = 1$ (D) $n = 4$, $l = 2$

Atomic Structure/Periodicity

113. Which pair consists of species that are isoelectronic?
 (A) Na^+ , K^+ (B) Cl , Cl^-
 (C) Fe^{2+} , Mn^{2+} (D) Ar , Ca^{2+}
114. In which series are the species listed in order of increasing size?
 (A) N , O , F (B) Na , Mg , K
 (C) Cr , Cr^{2+} , Cr^{3+} (D) Cl , Cl^- , S^{2-}
115. Which type of radiation has the highest frequency?
 (A) infrared (B) microwave
 (C) ultraviolet (D) X-ray
116. How many orbitals have the quantum numbers:
 $n = 4$, $\ell = 3$, $m_\ell = 0$
 (A) 7 (B) 3 (C) 1 (D) 0
117. What is the ground state electron configuration of a $_{27}\text{Co}$ atom in the gas phase?
 (A) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7$
 (B) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^9$
 (C) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^1$
 (D) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^2$
118. Which equation represents the first ionization of calcium?
 (A) $\text{Ca(s)} \rightarrow \text{Ca}^+(\text{g}) + \text{e}^-$
 (B) $\text{Ca(g)} \rightarrow \text{Ca}^+(\text{g}) + \text{e}^-$
 (C) $\text{Ca}^+(\text{g}) \rightarrow \text{Ca}^{2+}(\text{g}) + \text{e}^-$
 (D) $\text{Ca}^{2+}(\text{g}) + \text{e}^- \rightarrow \text{Ca}^+(\text{g})$
119. Electronegativities change both down a group and across a period. In general these changes are to:
- | down a group | across a period from left to right |
|--------------|------------------------------------|
| (A) increase | increase |
| (B) increase | decrease |
| (C) decrease | increase |
| (D) decrease | decrease |
120. When the atoms: Ba , Cs , Mg , Na are arranged in order of increasing size, what is the correct order?
 (A) $\text{Cs} < \text{Na} < \text{Mg} < \text{Ba}$ (B) $\text{Mg} < \text{Na} < \text{Ba} < \text{Cs}$
 (C) $\text{Mg} < \text{Ba} < \text{Na} < \text{Cs}$ (D) $\text{Ba} < \text{Mg} < \text{Na} < \text{Cs}$
121. What is the total number of p electrons in a single phosphorus atom in its ground state?
 (A) 3 (B) 5 (C) 9 (D) 15
122. Which element has the largest atomic radius?
 (A) Br (B) K (C) Mg (D) Na
123. Which pair of symbols identifies two elements that are metalloids?
 (A) B and Ge (B) Mg and Si
 (C) P and As (D) Ti and V
124. Which pair of symbols represents nuclei that have the same number of neutrons?
 (A) $^{56}_{26}\text{Fe}$ and $^{58}_{28}\text{Ni}$ (B) $^{58}_{26}\text{Fe}$ and $^{56}_{26}\text{Fe}^{2+}$
 (C) $^{57}_{27}\text{Co}$ and $^{57}_{28}\text{Ni}$ (D) $^{57}_{28}\text{Ni}$ and $^{58}_{28}\text{Ni}$
125. Green light has a wavelength that is slightly shorter than that of
 (A) gamma rays. (B) orange light.
 (C) violet light. (D) X-rays.
126. Which is the electron configuration for an Fe(III) ion in its ground state?
 (A) $[\text{Ar}] 3d^5$ (B) $[\text{Ar}] 3d^6$
 (C) $[\text{Ar}] 4s^2 3d^3$ (D) $[\text{Ar}] 4s^2 3d^6$
127. Which metal is the most reactive?
 (A) silver (B) lead
 (C) iron (D) cesium
128. Which element has the largest first ionization energy?
 (A) Li (B) B (C) N (D) Na
129. All of the energy levels listed are allowed EXCEPT
 (A) 3f. (B) 4d. (C) 5p. (D) 7s.
130. A sulfur atom has the electron configuration: $1s^2 2s^2 2p^6 3s^2 3p^4$. How many orbitals in this ground state atom are occupied by at least one electron?
 (A) 12 (B) 9 (C) 8 (D) 5

Atomic Structure/Periodicity

131. The shapes of s and p orbitals are determined by which quantum number(s)?
- | |
|------------|
| I. n |
| II. l |
| III. m_l |
- (A) I only (B) II only
(C) III only (D) II and III only
132. What are the elements called for which the 4f subshell is being filled?
- (A) transition metals (B) metalloids
(C) lanthanides (D) actinides
133. Which set of quantum numbers (n, l, m_l, m_s) is not permitted by the rules of quantum mechanics?
- (A) 1, 0, 0, $\frac{1}{2}$ (B) 2, 1, -1, $-\frac{1}{2}$
(C) 3, 3, 1, $-\frac{1}{2}$ (D) 4, 3, 2, $\frac{1}{2}$
134. How many unpaired electrons are in a gaseous iron atom in its ground state?
- (A) 6 (B) 4 (C) 2 (D) 0
135. Evidence for the electron arrangement in atoms has been obtained primarily from the study of
- (A) isotopes. (B) radioactivity.
(C) stoichiometry. (D) atomic spectra.
136. Which statement about the radii of atoms and their ions is correct?
- (A) Cations are smaller than their atoms, anions are larger.
(B) Cations and anions are both smaller than their atoms.
(C) Cations and anions are both larger than their atoms.
(D) Cations are larger than their atoms, anions are smaller.
137. When the atoms; Li, Be, B, Na, are arranged in order of increasing atomic radius which is the correct order?
- (A) Li, Be, B, Na (B) Li, Na, B, Be
(C) Na, Li, Be, B (D) B, Be, Li, Na
138. Which type of radioactive decay produces a daughter nucleus with a higher atomic number?
- (A) α (B) β^- (C) γ (D) β^+
139. The emission spectrum of hydrogen in the visible region consists of
- (A) a continuous band of light.
(B) a series of equally spaced lines.
(C) a series of lines that are closer at low energies.
(D) a series of lines that are closer at high energies.
140. Which atom in its ground state has the most unpaired electrons?
- (A) Ge (B) As (C) Se (D) Br
141. An monoatomic ion that has 18 electrons and a +2 charge
- (A) has 16 protons. (B) has the symbol Ar^{2+} .
(C) has 18 neutrons. (D) is isoelectronic with Ar.
142. Which atom has the largest atomic radius?
- (A) Li (B) K (C) As (D) Br
143. What is the maximum number of electrons that occupy the $n = 3$ level?
- (A) 6 (B) 8 (C) 10 (D) 18
144. How does the reducing ability of the elements vary across the period from Na to Ar? It
- (A) decreases steadily.
(B) increases steadily.
(C) decreases then increases.
(D) increases then decreases.
145. In a hydrogen atom, which transition produces a photon with the highest energy?
- (A) $n = 3 \rightarrow n = 1$ (B) $n = 5 \rightarrow n = 3$
(C) $n = 12 \rightarrow n = 10$ (D) $n = 22 \rightarrow n = 20$
146. How many orbitals in a ground state oxygen atom are completely filled?
- (A) 1 (B) 2 (C) 3 (D) 4
147. Which atom has the smallest first ionization energy?
- (A) Na (B) K (C) Mg (D) Ca
148. The electron configuration of a cobalt atom is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^2$.
How many unpaired electrons are present in a gaseous Co^{3+} ion in its ground state?
- (A) 6 (B) 4 (C) 2 (D) 0

Atomic Structure/Periodicity

149. When the atoms; P ($Z = 15$), S ($Z = 16$) and As ($Z = 33$), are arranged in order of increasing radius, what is the correct order?
- (A) P, S, As (B) As, S, P
(C) S, P, As (D) P, As, S
150. The oxide of which element is the most ionic?
- (A) Al (B) B (C) C (D) Si
151. Gaseous atoms of which of these elements contain one or more unpaired electrons?
- Ge ($Z = 32$) As ($Z = 33$) Se ($Z = 34$)
- (A) As only (B) Ge and As only
(C) Ge and Se only (D) Ge, As, and Se
152. How many orbitals contain one or more electrons in an isolated ground state iron atom ($Z = 26$)?
- (A) 13 (B) 14 (C) 15 (D) 16
153. Which property decreases from left to right across the periodic table and increases from top to bottom?
- (A) atomic radius (B) electronegativity
(C) ionization energy (D) melting point
154. According to the Aufbau principle, which is the sequential order of filling subshells in a ground state atom?
- (A) $3s3p3d$ (B) $3p4s3d$
(C) $3d4s4p$ (D) $4p4d4f$
155. Which has the highest ionization energy for the removal of the *second* electron?
- (A) F (B) Ne (C) Na (D) Mg
156. Which pair of elements have chemical properties that are the most similar?
- (A) Be and B (B) Al and Ga
(C) Co and Cu (D) F and I
157. When a phosphorus atom is converted to a phosphide ion, what happens to the number of unpaired electrons and the total number of electrons around the phosphorus?
- | | unpaired electrons | total electrons |
|-----|--------------------|------------------|
| (A) | increases | increases |
| (B) | decreases | increases |
| (C) | increases | remains the same |
| (D) | decreases | remains the same |
158. Which list includes species with the same number of electrons?
- (A) Mg, Ca, Sr (B) Li^+ , Na^+ , K^+
(C) K^+ , Cl^- , S^{2-} (D) Fe^{2+} , Co^{2+} , Ni^{2+}
159. Which pair of symbols represents nuclei with the same number of neutrons?
- (A) ^{56}Co and ^{58}Co (B) ^{57}Mn and ^{57}Fe
(C) ^{57}Fe and ^{58}Ni (D) ^{57}Co and ^{58}Ni
160. Which list includes elements in order of *increasing* metallic character?
- (A) Si, P, S (B) As, P, N
(C) Al, Ge, Sb (D) Br, Se, As
161. How many unpaired electrons are in an iron atom in its ground state?
- (A) 6 (B) 4 (C) 2 (D) 0
162. How do the energy gaps between successive electron energy levels in an atom vary from low to high n values?
- (A) All energy gaps are the same.
(B) The energy gap decreases as n increases.
(C) The energy gap increases as n increases.
(D) The energy gap changes unpredictably as n increases.
163. Which element has an outer electron configuration of s^2p^4 ?
- (A) Ca (B) Cr (C) Ge (D) Se
164. How many unpaired electrons does a gaseous atom of phosphorus, P, have in its ground state?
- (A) 1 (B) 3 (C) 5 (D) 7
165. Which element has the lowest first ionization energy?
- (A) B (B) C (C) Al (D) Si
166. Which of these elements has the greatest electronegativity?
- (A) Br (B) N (C) O (D) S
167. Which oxide produces the most acidic solution when 0.1 mol is added to 1 L of H_2O ?
- (A) BaO (B) BaO_2 (C) SO_2 (D) SO_3